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United States Patent

12408347

Kind Code

B2

Date of Patent

September 02, 2025

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Method for forming a 3-D semiconductor memory structure comprising horizontal and vertical conductive lines

Abstract

A method for forming a semiconductor memory structure is provided. The method includes forming a stack over a substrate, and the stack includes first dielectric layers and second dielectric layers vertically alternately arranged. The method also includes forming first dielectric pillars through the stack, and etching the stack to form first trenches. Sidewalls of the first dielectric pillars are exposed from the first trenches. The method also includes removing the first dielectric pillars to form through holes, removing the second dielectric layers of the stack to form gaps between the first dielectric layers, and forming first conductive lines in the gaps.

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Appl. No.: 17/667873

Filed: February 09, 2022

Prior Publication Data

Document Identifier

Publication Date

US 20230024339 A1

Jan. 26, 2023

Related U.S. Application Data

Publication Classification

Int. Cl.: **H10B51/20** (20230101); **H01L23/48** (20060101); **H01L23/528** (20060101); **H10B51/10** (20230101)

U.S. Cl.:

CPC **H10B51/20** (20230201); **H01L23/481** (20130101); **H01L23/5283** (20130101); **H10B51/10** (20230201);

Field of Classification Search

CPC: H10B (51/00); H10B (51/20); H01L (23/48); H01L (23/481); H01L (23/528); H01L (23/5283)

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Background/Summary

PRIORITY CLAIM (1) This application claims the benefit of U.S. Provisional Application No. 63/224,113, filed on Jul. 21, 2021 and entitled “SEMICONDUCTOR MEMORY DEVICE WITH AN STRUT STRUCTURE AND METHOD FOR FORMING THE SAME,” which is incorporated herein by reference in its entirety.

BACKGROUND

(1) The semiconductor integrated circuit (IC) industry has experienced exponential growth. Technological advances in IC materials and design have produced generations of ICs where each generation has smaller and more complex circuits than the previous generation. In the course of IC evolution, functional density (i.e., the number of interconnected devices per chip area) has generally increased while geometric size (i.e., the smallest component (or line) that can be created using a fabrication process) has decreased. This scaling down process generally provides benefits by increasing production efficiency and lowering associated costs. Such scaling down has also increased the complexity of processing and manufacturing ICs and, for these advancements to be realized, similar developments in IC processing and manufacturing are needed.

(2) One type of device targeted for increased capacity and integration is a memory device. Two-dimensional (2D) memory arrays are prevalent in electronic devices and may include, for example, NOR flash memory arrays, NAND flash memory arrays, dynamic random-access memory (DRAM) arrays, and so on. However, 2D memory arrays are reaching scaling limits and are hence reaching limits on memory density. Three-dimensional (3D) memory arrays are a promising candidate for increasing memory density and may include, for example, 3D NAND flash memory arrays, 3D NOR flash memory arrays, and so on.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying Figures. It should be noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

(2) FIGS. 1A, 1B, 1C, 1D, 1E, 1F, 1G, 1H, 1I, 1J, 1K, 1L, 1M and 1N are perspective views illustrating the formation of a semiconductor memory structure, in accordance with some embodiments of the disclosure.

(3) FIGS. 1B-1, 1C-1, 1D-1, 1E-1, 1F-1, 1G-1, 1H-1, 1I-1, 1J-1, 1K-1, 1L-1, 1M-1 and 1N-1 are plan views illustrating the formation of a semiconductor memory structure, in accordance with some embodiments of the disclosure.

(4) FIGS. 1G-2, 1H-2, 1H-2, 1K-2, 1M-2 and 1N-2 are cross-sectional views taken along lines I-I shown in FIGS. 1G-1, 1H-1, 1I-1, 1K-1, 1M-1 and 1N-1, respectively, in accordance with some embodiments of the disclosure.

(5) FIGS. 1H-3, 1K-3, 1M-3 and 1N-3 are cross-sectional views taken along lines II-II shown in FIGS. 1H-1, 1K-1, 1M-1 and 1N-1, respectively, in accordance with some embodiments of the disclosure.

(6) FIG. 2 is a cross-sectional view of a semiconductor memory structure, in accordance with some embodiments of the disclosure.

(7) FIGS. 3A and 3B are cross-sectional views illustrating the formation of a semiconductor memory structure, in accordance with some embodiments of the disclosure.

(8) FIGS. 4A and 4B are cross-sectional views illustrating the formation of a semiconductor memory structure, in accordance with some embodiments of the disclosure.

DETAILED DESCRIPTION

(9) The following disclosure provides many different embodiments, or examples, for implementing different features of the subject matter provided. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

(10) Some variations of the embodiments are described. Throughout the various views and illustrative embodiments, like reference numerals are used to designate like elements. It should be understood that additional operations can be provided before, during, and after the method, and some of the operations described can be replaced or eliminated for other embodiments of the method.

(11) Furthermore, when a number or a range of numbers is described with “about,” “approximate,” and the like, the term is intended to encompass numbers that are within a reasonable range including the number described, such as within $\pm 10\%$ of the number described or other values as understood by a person skilled in the art. For example, the term “about 5 nm” encompasses the dimension range from 4.5 nm to 5.5 nm.

(12) 3D NOR memory is a flash memory in which memory cells are stacked vertically to provide much higher storage density and lower cost per gigabyte in comparison to an existing memory. The density of 3D memory is increased by stacking more memory gate films for forming word lines or transistor layers. The stack of the gate films is vertically cut through into several strips and trenches between the strips. For example, the aspect ratio (height/width) of the trenches can be greater than about 20. However, as the strips including the gate films become narrower and taller, the risk that the strips collapse and/or wiggle is higher, thereby decreasing the manufacturing yield of the memory device.

(13) In addition, dielectric layers of the strips are replaced with the gate films by etching-deposition-etching back processes. The gate films usually suffer lateral recess and have concave etched surfaces, which may lead to higher resistance. Furthermore, channel layers formed along the strips may also have uneven profiles (also referred to as bird's beak issue), thereby degrading the performance of the memory device.

(14) Embodiments for forming a semiconductor memory structure are provided. The method for forming the semiconductor memory structure may include forming a stack which includes first dielectric layers and second dielectric layers alternately arranged, forming first dielectric pillars through the stack, and etching the stack to form a plurality of first trenches and strips between the trenches, in accordance with some embodiments. The first dielectric pillars may support the strips, thereby decreasing the risk of collapsing and/or wiggling of the strips, in accordance with some embodiments. Therefore, the manufacturing yield of the resulting semiconductor memory device may improve.

(15) In addition, the method also includes forming sacrificial layers in the trenches, replacing the second dielectric layers with conductive lines, and forming channel layers along strips including the conductive lines and the first dielectric layers. As a result, the conductive lines may be formed with substantially flat sidewalls, and the channel layers formed thereon also have substantially flat profiles. Therefore, the performance of the resulting semiconductor memory device may improve.

(16) FIGS. 1A, 1B, 1C, 1D, 1E, 1F, 1G, 1H, 1I, 1J, 1K, 1L, 1M and 1N are perspective views illustrating the formation of a semiconductor memory structure **100**, in accordance with some embodiments. FIGS. 1B-1, 1C-1, 1D-1, 1E-1, 1F-1, 1G-1, 1H-1, 1I-1, 1J-1, 1K-1, 1L-1, 1M-1 and

1N-1 are plan views of the semiconductor memory structure **100** that are horizontally cut through a second dielectric layer **106** or first conductive lines **122**, in accordance with some embodiments.

(17) FIG. **1A** illustrate a semiconductor memory structure **100**, in accordance with some embodiments. The semiconductor memory structure **100** includes a substrate **102**, in accordance with some embodiments. In some embodiments, the substrate **102** is a semiconductor substrate such as a silicon substrate. In some embodiments, the substrate **102** includes an elementary semiconductor such as germanium; a compound semiconductor such as gallium nitride (GaN), silicon carbide (SiC), gallium arsenide (GaAs), gallium phosphide (GaP), indium phosphide (InP), indium arsenide (InAs), and/or indium antimonide (InSb); an alloy semiconductor such as SiGe, GaAsP, AlInAs, AlGaAs, GaInAs, GaInP, and/or GaInAsP; or a combination thereof.

(18) In some embodiments, the substrate **102** includes semiconductor devices formed on the semiconductor substrate. For example, the semiconductor device may be peripheral circuits which may include various devices such as metal-oxide-semiconductor (MOS) FETs, fin FETs, nanostructure FETs (e.g., gate-all-around (GAA) FETs), or another suitable type of semiconductor device.

(19) In some embodiment, the substrate **102** may also include an interconnect structure that includes multiple dielectric layers and electrically conductive features (such as contacts, metal lines and/or conductive vias) in the dielectric layers. The peripheral circuits may be operable to access and/or control devices of a memory cell array (e.g., to perform read/write/erase operations) formed thereabove through the conductive features of the interconnect structure.

(20) For a better understanding of the semiconductor memory structure **100**, X-Y-Z coordinate reference is provided in the figures of the present disclosure. X-axis and Y-axis are generally orientated along the lateral (or horizontal) directions that are parallel to the main surface of the substrate **102**. Y-axis is transverse (e.g., substantially perpendicular) to the X-axis. Z-axis is generally oriented along the vertical direction that is perpendicular to the main surface of the substrate **102** (or the X-Y plane).

(21) A stack including first dielectric layers **104** and second dielectric layers **106** is formed over the substrate **102**, as shown in FIG. **1A**, in accordance with some embodiments. In some embodiments, the semiconductor memory structure **100** may include various device regions, e.g., a logic region, a memory cell array region, an analog region, a peripheral region, another suitable region, and/or a combination thereof. The stack is formed in the memory cell array region of the semiconductor memory structure **100**, in accordance with some embodiments.

(22) The first dielectric layers **104** and the second dielectric layers **106** are vertically alternately stacked, in accordance with some embodiments. In some embodiments, the second dielectric layers **106** are configured as sacrificial layers which will be replaced with conductive lines (such as word lines). In some embodiments, the first dielectric layers **104** are configured as insulating layers to physically and electrically isolate conductive lines from one another.

(23) In some embodiments, the number of the first dielectric layers **104** is one more than the number of the second dielectric layers **106**. That is, both the top layer and the bottom layer of the stack are first dielectric layers **104**. Although five first dielectric layers **104** and four second dielectric layers **106** are shown in FIG. **1A**, the numbers of the first dielectric layers **104** and the second dielectric layers **106** are not limited thereto, and can be in a range from 2 to about 100.

(24) In some embodiments, the thickness of each of the first dielectric layers **104** is in a range from about 10 nm to about 200 nm. In some embodiments, the uppermost first dielectric layer **104** is thicker than other first dielectric layers **104**. In some embodiments, the thickness of each of the second dielectric layers **106** is in a range from about 10 nm to about 200 nm.

(25) In some embodiments, the first dielectric layers **104** and the second dielectric layers **106** are made dielectric materials such as silicon oxide (SiO₂), silicon oxynitride (SiON), silicon nitride (SiN), silicon oxycarbide (SiOC), silicon carbon nitride (SiCN), silicon oxycarbonitride (SiOCN), or a combination thereof. In some embodiments, the first dielectric layers **104** and the

second dielectric layers **106** are made of different materials and have a difference in etching selectivity. In some embodiments, the first dielectric layers **104** are made of an oxide-based dielectric material (such as silicon oxide) and the second dielectric layers **106** are made of a nitride-based dielectric material (such as silicon nitride).

(26) In some embodiments, the first dielectric layers **104** and the second dielectric layers **106** are deposited using atomic layer deposition (ALD), chemical vapor deposition (CVD) (such as low-pressure CVD (LPCVD), plasma-enhanced CVD (PECVD), high density plasma CVD (HDP-CVD)), another suitable technique, and/or a combination thereof. In some embodiments, the formation of the stack may be integrated into the CMOS manufacturing process, e.g., the back end of line (BEOL) process. For example, the stack may be located at the fifth level (M5) and/or sixth metal layer (M6) of metal layers of an interconnect structure.

(27) FIGS. **1B** and **1B-1** illustrate the formation of first through holes **108**, in accordance with some embodiments.

(28) First through holes **108** are formed through the stack including the first dielectric layers **104** and the second dielectric layers **106**, as shown in FIGS. **1B** and **1B-1**, in accordance with some embodiments. In some embodiments, the formation of the first through holes **108** includes forming a patterned mask layer (not shown) having opening patterns corresponding to the first through holes **108** over the stack, and then etching the stack using the patterned mask layer to transfer the opening patterns into the stack until the substrate **102** is exposed. In some embodiments, the patterned mask layer is a patterned photoresist layer which is formed by a photolithography process. The etching process may be an anisotropic etching process such as dry plasma etching, an isotropic etching process such as dry chemical etching, remote plasma etching, wet chemical etching, and/or a combination thereof.

(29) In some alternative embodiments, the patterned mask layer is a patterned hard mask layer, which is formed by depositing a dielectric layer, forming a patterned photoresist layer over the dielectric layer, and etching the dielectric layer to transfer the opening patterns of the photoresist layer into the dielectric layer. The patterned mask layer may be removed during the etching process or by an additional process (such as etching, wet strip and/or ashing).

(30) The first through holes **108** are arranged in a row/column configuration, in accordance with some embodiments. For example, rows of first through holes **108** extend in the X direction, and columns of first through holes **108** extend in the Y direction. In some embodiments, the first through holes **108** in neighboring two columns may be staggered with each other (e.g., not overlap in the X direction).

(31) In some embodiments, the first through holes **108** have a dimension D1 as measured in the X direction. In some embodiments, the dimension D1 is in a range from about 50 nm to about 300 nm. In some embodiments, the first through holes **108** have a dimension D2 as measured in the Y direction. In some embodiments, the dimension D2 is in a range from about 50 nm to about 500 nm.

(32) In some embodiments, the first through holes **108** are arranged at an X-pitch P.sub.X (in the X direction), which is in a range from about 150 nm to about 500 nm. In some embodiments, the first through holes **108** are arranged at a Y-pitch P.sub.Y (in the Y direction), which is in a range from about 500 nm to about 10 μ m. In some embodiments, the ratio (P.sub.Y/D2) of the Y-pitch P.sub.Y to the dimension D2 is in a range from about 1 to about 50.

(33) FIGS. **1C** and **1C-1** illustrate the formation of first dielectric pillars **110**, in accordance with some embodiments.

(34) First dielectric pillars **110** are formed in the first through holes **108**, as shown in FIGS. **1C** and **1C-1**, in accordance with some embodiments. The first dielectric pillars **110** penetrate through the stack and are in contact with the substrate **102**, in accordance with some embodiments. The first dielectric pillars **110** are configured to support subsequently formed strips from collapsing and/or wiggling, in accordance with some embodiments.

(35) In some embodiments, the first dielectric pillars **110** are made of a dielectric material such as silicon nitride (SiN), silicon oxynitride (SiON), silicon oxide (SiO₂), silicon oxycarbide (SiOC), silicon carbon nitride (SiCN), silicon oxycarbonitride (SiOCN), or a combination thereof. In some embodiments, the first dielectric pillars **110** are made of the same material as the second dielectric layers **106**, e.g., a nitride-based dielectric material (such as silicon nitride).

(36) In some embodiments, the first dielectric pillars **110** are formed by depositing a dielectric material to overfill the first through holes **108**, and then planarizing the dielectric material to removing the portion of the dielectric material formed above the stack until the uppermost first dielectric layer **104** is exposed. The deposition process may be ALD, CVD (such as LPCVD, PECVD, HDP-CVD, high aspect ratio process (HARP), flowable CVD (FCVD)), another suitable technique, and/or a combination thereof. The planarizing process may be an etching back process and/or chemical mechanical polishing (CMP).

(37) The first dielectric pillars **110** are arranged in a row/column configuration, in accordance with some embodiments. For example, rows of first dielectric pillars **110** extend in the X direction, and columns of first dielectric pillars **110** extend in the Y direction. In some embodiments, the first dielectric pillars **110** in neighboring two columns may be staggered with each other (e.g., not overlap in the X direction).

(38) In some embodiments, the first dielectric pillars **110** also have the dimension D1 as measured in the X direction. In some embodiments, the dimension D1 is in a range from about 50 nm to about 300 nm. In some embodiments, the first dielectric pillars **110** also have the dimension D2 as measured in the Y direction. In some embodiments, the dimension D2 is in a range from about 50 nm to about 500 nm.

(39) In some embodiments, the first dielectric pillars **110** are arranged at an X-pitch P_X (in the X direction), which is in a range from about 150 nm to about 500 nm. In some embodiments, the first dielectric pillars **110** are arranged at a Y-pitch P_Y (in the Y direction), which is in a range from about 500 nm to about 10 μm. In some embodiments, the ratio (P_Y/D2) of the Y-pitch P_Y to the dimension D2 is in a range from about 1 to about 50. If the ratio (P_Y/D2) or the Y-pitch P_Y is too small, the areal density of memory cells may decrease. If the ratio (P_Y/D2) or the Y-pitch P_Y is too large, the risk that subsequently formed strips collapse and/or wiggle may increase.

(40) FIGS. 1D and 1D-1 illustrate the formation of first trenches **112**, in accordance with some embodiments.

(41) First trenches **112** are formed through the stack including first dielectric layers **104** and the second dielectric layers **106**, as shown in FIGS. 1D and 1D-1, in accordance with some embodiments. In some embodiments, the first trenches **112** extend in the Y direction and are formed aligned with the columns of first dielectric pillars **110**. In some embodiments, each of the first trenches **112** is formed between and exposes neighboring two first dielectric pillars **110** in a column.

(42) In some embodiments, the formation of the first trenches **112** includes forming a patterned mask layer (not shown) having trench patterns corresponding to the first trenches **112** over the stack, and then etching the stack using the patterned mask layer to transfer the trench patterns into the stack until the substrate **102** is exposed. The stack is cut through into several strips **114**, which protrude from between the first trenches **112**, in accordance with some embodiments. In some embodiments, the patterned mask layer is a patterned photoresist layer which is formed by a photolithography process. The etching process may be an anisotropic etching process such as dry plasma etching, an isotropic etching process such as dry chemical etching, remote plasma etching, wet chemical etching, and/or a combination thereof.

(43) Each of the first dielectric pillars **110** may support the neighboring two strips **114**, thereby decreasing the risk of collapsing and/or wiggling of the strips **114**, in accordance with some embodiments. As a result, the manufacturing yield of the resulting semiconductor memory device

may improve.

(44) In some alternative embodiments, the patterned mask layer is a patterned hard mask layer, which is formed by depositing a dielectric layer, forming a patterned photoresist layer over the dielectric layer, and etching the dielectric layer to transfer the trench patterns of the photoresist layer into the dielectric layer. The patterned mask layer may be removed during the etching process or by an additional process (such as etching, wet strip and/or ashing).

(45) In some embodiments, the strips include alternatively stacked the first dielectric layers **104** and the second dielectric layers **106**. In some embodiments, the strips **114** extend in the Y direction and are arranged in parallel in the X direction. That is, the strips **114** have longitudinal axes parallel to Y direction. In some embodiments, the strips **114** have a dimension D3 as measured in the X direction. In some embodiments, the dimension D3 is in a range from about 50 nm to about 250 nm. In some embodiments, the strips **114** (or the first trenches **112**) also have X-pitch P.sub.X (in the X direction), which is in a range from about 150 nm to about 500 nm.

(46) FIGS. **1E** and **1E-1** illustrate the formation of sacrificial layers **116**, in accordance with some embodiments.

(47) Sacrificial layers **116** are formed in the first trenches **112**, as shown in FIGS. **1E** and **1E-1**, in accordance with some embodiments. Each of the sacrificial layers **116** is formed between neighboring two first dielectric pillars **110** in a column and between neighboring two strips **114**, in accordance with some embodiments. The sacrificial layers **116** are in contact with the first dielectric pillars **110**, the strips **114**, and the substrate **102**, in accordance with some embodiments. The sacrificial layers **116** are configured as retaining walls to constrain subsequently formed conductive lines to have sidewalls with desired profiles, in accordance with some embodiments.

(48) In some embodiments, the sacrificial layers **116** are made of a semiconductor material such as silicon (Si), germanium (Ge) and/or silicon germanium (SiGe), and/or a metal oxide such as zirconium dioxide (ZrO.sub.2), hafnium oxide (HfO.sub.2), aluminum oxide (Al.sub.2O.sub.3), yttrium oxide (Y.sub.2O.sub.3), lanthanum oxide (La.sub.2O.sub.3), or a combination thereof. In some embodiments, the sacrificial layers **116** have a different etching selectivity than the first dielectric layers **104**, the second dielectric layers **106**, the first dielectric pillars **110**, and subsequently formed conductive lines.

(49) In some embodiments, the sacrificial layers **116** are formed by depositing a material for the sacrificial layers **116** to overfill the first trenches **112**, and then planarizing the material for the sacrificial layers **116** to remove the portion of the material for the sacrificial layers **116** formed above the strips **114** until the uppermost first dielectric layer **104** is exposed. The deposition process may be ALD, CVD (such as LPCVD, PECVD, HDP-CVD, HARP, and FCVD), another suitable technique, and/or a combination thereof. The planarizing process may be an etching back process and/or chemical mechanical polishing (CMP).

(50) FIGS. **1F** and **1F-1** illustrate the formation of second through holes **118**, in accordance with some embodiments.

(51) Second through holes **118** are formed through the stack by removing the first dielectric pillars **110**, as shown in FIGS. **1F** and **1F-1**, in accordance with some embodiments. The removal process may be an isotropic etching process such as wet chemical etching, dry chemical etching or remote plasma etching, an anisotropic etching process such as dry plasma etching, and/or a combination thereof. The second through holes **118** expose the sacrificial layers **116**, the strips **114** and the substrate **102**, in accordance with some embodiments.

(52) The second through holes **118** are arranged in a row/column configuration, in accordance with some embodiments. For example, rows of second through holes **118** extend in the X direction, and columns of second through holes **118** extend in the Y direction. In some embodiments, the second through holes **118** in neighboring two columns may be staggered with each other (e.g., not overlap in the X direction).

(53) In some embodiments, the second through holes **118** also have the dimension D1 as measured

in the X direction. In some embodiments, the dimension D1 is in a range from about 50 nm to about 300 nm. In some embodiments, the second through holes **118** also have a dimension D2 as measured in the Y direction. In some embodiments, the dimension D2 is in a range from about 50 nm to about 500 nm.

(54) In some embodiments, the second through holes **118** have X-pitch P.sub.X (in the X direction), which is in a range from about 500 nm to about 10 μ m. In some embodiments, the second through holes **118** have Y-pitch P.sub.Y (in the Y direction), which is in a range from about . . . nm to about . . . nm. In some embodiments, the ratio of the Y-pitch P.sub.Y to the dimension D2 (P.sub.Y/D2) is in a range from about 1 to about 50.

(55) FIGS. **1G** and **1G-1** illustrate the formation of gaps **120**, in accordance with some embodiments.

(56) An etching process is performed to removing the second dielectric layers **106** of the strips **114**, thereby forming gaps **120**, as shown in FIGS. **1G** and **1G-1**, in accordance with some embodiments. The gaps **120** are defined by the first dielectric layers **104** of the strips **114** and the sacrificial layers **116**, in accordance with some embodiments. The etching process may be an isotropic etching process such as wet chemical etching, dry chemical etching or remote plasma etching, and/or a combination thereof.

(57) During the etching process, an etchant is introduced into the second through holes **118** to laterally etch away the second dielectric layers (**106** of FIG. **1F**), thereby forming the gaps **120**, in accordance with some embodiments. The first dielectric layers **104** and the sacrificial layers **116** have different etching selectivity than the second dielectric layers in the etching process, and thus the first dielectric layers **104** and the sacrificial layers **116** remain substantially unetched.

(58) FIG. **1G-2** is a cross-sectional view taken along line I-I shown in FIG. **1G-1**, in accordance with some embodiments. The gap **120** also has a dimension D3 as measured in the X direction. In some embodiments, the dimension D3 is in a range from about 50 nm to about 250 nm. The gap **120** has a dimension D4 (substantially the same as the thickness of the second dielectric layer **106**) as measured in the Z direction. In some embodiments, the dimension D4 is in a range from about 10 nm to about 200 nm. In some embodiments, the dimension D3 and the dimension D4 of the gap **120** are less than the dimension D1 of the second through hole **118**.

(59) FIGS. **1H** and **1H-1** illustrate the formation of first conductive lines **122**, in accordance with some embodiments.

(60) The first conductive lines **122** are formed in the gaps (**120** of FIG. **1G**), as shown in FIGS. **1H** and **1H-1**, in accordance with some embodiments. As a result, the second dielectric layers **106** of the strips are replaced with the first conductive lines **122**, in accordance with some embodiments. The strips after replacement are denoted as strips **114'**. The first conductive lines **122** extend in the Y direction and are separated from one another in the Z direction and the X direction, in accordance with some embodiments. The first conductive lines **122** are in contact with the first dielectric layers **104** and the sacrificial layers **116**, in accordance with some embodiments. In some embodiments, the first conductive lines **122** are gate electrodes of memory transistors and function as word lines of the resulting semiconductor memory device, such as NOR-type flash memory.

(61) FIGS. **1H-2** and **1H-3** are cross-sectional views taken along lines I-I and II-II shown in FIG. **1H-1** and illustrate the details of the formation of the first conductive lines **122**, in accordance with some embodiments. In some embodiments, the first conductive lines **122** are made of one or more electrically conductive materials. In some embodiments, the first conductive lines **122** have a multi-layer structure including, for example, a barrier/adhesive layer, a metal bulk layer, another suitable layer, and/or a combination thereof.

(62) For example, a barrier layer **124** is first deposited along the surfaces of the first dielectric layers **104** and the sacrificial layers **116** exposed from the gaps (**120** of FIG. **1F**) and the second through hole **118** to partially fill the gaps (**120** of FIG. **1F**) and the second through hole **118**, in accordance with some embodiments. The barrier layer **124** is used to prevent the metal from the

subsequently formed metal material from diffusing into the dielectric material (e.g., the first dielectric layers **104**).

(63) The barrier layer **124** may be made of titanium nitride (TiN), tantalum (Ta), tantalum nitride (Ta₂N₃), titanium (Ti), cobalt tungsten (CoW), another suitable material, and/or a combination thereof. If the subsequently formed metal material does not easily diffuse into the dielectric material, the barrier layer may be omitted. The deposition process may be ALD, CVD, PVD, e-beam evaporation, ECP, ELD, another suitable method, or a combination thereof.

(64) A metal bulk layer **126** is then deposited on the barrier layer **124**, in accordance with some embodiments. In some embodiments, because the dimension D3 and the dimension D4 (FIG. 1G-2) of the gap (**120** of FIG. 1F) are less than the dimension D1 (FIG. 1G-2) of the second through hole **118**, the deposition process can be controlled such that the metal bulk layer **126** overfills the gaps **120** and partially fills the second through holes **118**.

(65) In some embodiments, the metal bulk layer **126** is made of conductive material with low resistance and good gap-fill ability, for example, tungsten (W), cobalt (Co), ruthenium (Ru), copper (Cu), aluminum (Al), nickel (Ni), titanium (Ti), tantalum (Ta), molybdenum (Mo), another suitable metal material, and/or a combination thereof. The deposition process may be ALD, CVD, PVD, e-beam evaporation, ECP, ELD, another suitable method, or a combination thereof.

(66) After the deposition of the metal bulk layer **126**, an etching back process is performed to remove portions of the barrier layer **124** and the metal bulk layer **126** outside the gaps (**120** of FIG. 1F). The etching process may be an isotropic etching process such as wet chemical etching, dry chemical etching or remote plasma etching, and/or a combination thereof. Because the second through holes **118** are partially filled, during the etching process, an etchant is introduced into the remainders of the second through holes **118** to laterally etch away the barrier layer **124** and the metal bulk layer **126** until the sidewalls of the first dielectric layers **104** originally covered by the barrier layer **124** and the metal bulk layer **126** are exposed from the second through holes **118** again, as shown in FIG. 1H-2, in accordance with some embodiments. The portions of the barrier layer **124** and the metal bulk layer **126** within the gaps (**120** of FIG. 1F) in combination serve as the first conductive lines **122**, in accordance with some embodiments.

(67) In some embodiments, the first conductive lines **122** also have the maximum dimension D3 as measured in the X direction. In some embodiments, the dimension D3 is in a range from about 150 nm to about 500 nm. The first conductive lines **122** have sidewalls **122B** which are interfaced with (or covered by) the sacrificial layers **116**, and thus the sidewalls **122B** remain substantially flat after the etching process, as shown in FIGS. 1H-1, 1H-2 and 1H-2, in accordance with some embodiments. Channel layers will be formed on the substantially flat sidewalls **122B** of the first conductive lines **122**, in accordance with some embodiments.

(68) In some embodiments, due to the characteristics of the etching process, in the etching back process, the portions of the first conductive lines **122** exposed from the second through holes **118** may be laterally recessed, e.g., by a maximum depth D5, thereby forming notches **119**, as shown in FIG. 1H-2. In some embodiments, the depth D5 is less than 50 nm, e.g., in a range from about 0 nm to about 50 nm. In some embodiments, a ratio (D5/D3) of depth D5 of the notches **119** to the dimension D3 of the first conductive lines **122** is in a range from about 0.5 to about 0.95.

(69) The sidewalls of the first conductive lines **122** have concave portions **122A** that are exposed from the second through holes **118**, as shown in FIGS. 1H-1 and 1H-2. Channel layers would not be formed on the concave portion **122A** of the sidewalls of the first conductive lines **122**, in accordance with some embodiments.

(70) Therefore, by forming the sacrificial layers **116** and the second through holes **118**, the first conductive lines **122** may be formed with sidewalls, most of which are substantially flat and a small part of which are concave. As a result, the first conductive lines **122** may be formed with a greater cross-sectional area (in the X-Y plane), thereby improving the performance (e.g., lower resistance) of the resulting semiconductor memory device.

(71) FIGS. 1I and 1I-1 illustrate the formation of second dielectric pillars **128**, in accordance with some embodiments.

(72) Second dielectric pillars **128** are formed in the second through holes (**118** of FIG. 1H), as shown in FIGS. 1I and 1I-1, in accordance with some embodiments. The second dielectric pillars **128** are in contact with the first dielectric layers **104** and the first conductive lines **122** of the strips **114'**, the sacrificial layers **116** and the substrate **102**, in accordance with some embodiments. The second dielectric pillars **128** are configured to support strips **114'** from collapsing and/or wiggling in and after a subsequent etching process, in accordance with some embodiments.

(73) In some embodiments, the second dielectric pillars **128** are made of dielectric material such as silicon oxide (SiO_2), silicon oxynitride (SiON), silicon nitride (SiN), silicon oxycarbide (SiOC), silicon carbon nitride (SiCN), silicon oxycarbonitride (SiOCN), or a combination thereof. In some embodiments, the second dielectric pillars **128** are made of the same material as the first dielectric layers **104**, e.g., an oxide-based dielectric material (such as silicon oxide).

(74) In some embodiments, the second dielectric pillars **128** are formed by depositing a dielectric material to overfill the second through holes (**118** of FIG. 1H), and then planarizing the dielectric material to remove the portion of the dielectric material formed above the strips **114'** until the sacrificial layers **116** are exposed. The deposition process may be ALD, CVD (such as LPCVD, PECVD, HDP-CVD, HARP, and FCVD), another suitable technique, and/or a combination thereof. The planarizing process may be an etching back process and/or chemical mechanical polishing (CMP).

(75) The second dielectric pillars **128** are arranged in a row/column configuration, in accordance with some embodiments. For example, rows of second dielectric pillars **128** extend in the X direction, and columns of second dielectric pillars **128** extend in the Y direction. In some embodiments, the second dielectric pillars **128** in neighboring two columns may be staggered with each other (e.g., not overlap in the X direction).

(76) In some embodiments, the second dielectric pillars **128** have the maximum dimension D_r as measured in the X direction. In some embodiments, the dimension D_1' is in a range from about 50 nm to about 300 nm. In some embodiments, the second dielectric pillars **128** also have the dimension D_2 as measured in the Y direction. In some embodiments, the dimension D_2 is in a range from about 50 nm to about 500 nm.

(77) In some embodiments, the second dielectric pillars **128** are arranged at an X-pitch $P_{\text{sub.X}}$ (in the X direction), which is in a range from about 150 nm to about 500 nm. In some embodiments, the second dielectric pillars **128** are arranged at a Y-pitch $P_{\text{sub.Y}}$ (in the Y direction), which is in a range from about 500 nm to about 10 μm . In some embodiments, the ratio of the Y-pitch $P_{\text{sub.Y}}$ to the dimension D_2 ($P_{\text{sub.Y}}/D_2$) is in a range from about 1 to about 50. If the ratio ($P_{\text{sub.Y}}/D_2$) or the Y-pitch $P_{\text{sub.Y}}$ is too small, the areal density of memory cells may decrease. If the ratio ($P_{\text{sub.Y}}/D_2$) or the Y-pitch $P_{\text{sub.Y}}$ is too large, the risk that the strips **114'** including the first conductive lines **122** collapse and/or wiggle may increase.

(78) FIG. 1I-2 is a cross-sectional view taken along lines I-I shown in FIG. 1I-1, in accordance with some embodiments. The second dielectric pillars **128** include protruding portions filled in the notches (of **119** FIG. 1H-2), in accordance with some embodiments. The protruding portions of the second dielectric pillars **128** have convex side surfaces **128A** that mate and are interfaced with the concave portions **122A** of the sidewalls of the first conductive lines **122**, as shown in FIG. 1I-2, in accordance with some embodiments. In some embodiments, the convex side surfaces **128A** are in contact with both the barrier layer **124** and the metal bulk layer **126** of the first conductive lines **122**.

(79) The second dielectric pillars **128** also have substantially flat side surfaces **128B** interfaced with the first dielectric layers **104**, in accordance with some embodiments. The convex side surfaces **128A** and the substantially flat side surfaces **128B** are alternately arranged, in accordance with some embodiments.

(80) FIGS. 1J and 1J-1 illustrate the removal of the sacrificial layers **116**, in accordance with some embodiments.

(81) An etching process is performed to remove the sacrificial layers (**116** of FIG. 1I), thereby forming second trenches **130**, as shown in FIGS. 1J and 1J-1, in accordance with some embodiments. The etching process may be an anisotropic etching process such as dry plasma etching, an isotropic etching process such as dry chemical etching, remote plasma etching, wet chemical etching, and/or a combination thereof.

(82) Each of the second dielectric pillars **128** may support the neighboring two strips **114'**, thereby decreasing the risk of collapsing and/or wiggling of the strips **114'**, in accordance with some embodiments. As a result, the manufacturing yield of the resulting semiconductor memory device may improve.

(83) In some embodiments, the second trenches **130** extend in the Y direction and are formed aligned with the columns of second dielectric pillars **128**. In some embodiments, each of the second trenches **130** is formed between and exposes neighboring two second dielectric pillars **128** in a column. In some embodiments, the second trenches **130** also expose the first dielectric layers **104** and the first conductive lines **122** of the strips **114'**.

(84) FIGS. 1K and 1K-1 illustrate the formation of ferroelectric (FE) layers **132**, channel layers **134**, insulating layers **136** and capping layer **138**, in accordance with some embodiments.

(85) Ferroelectric layers **132**, channel layers **134**, insulating layers **136** and capping layers **138** are sequentially deposited over the semiconductor memory structure **100** to partially fill the second trenches (**130** of FIG. 1J), as shown in FIGS. 1K and 1K-1, in accordance with some embodiments. The remaining portions of the second trenches (**130** of FIG. 1J) are denoted as **131**.

(86) In some embodiments, the ferroelectric layers **132** are made of ferroelectric material such as hafnium oxide (HfO), zirconium oxide (ZrO), hafnium zirconium oxide (HfZrO), another suitable material, or a combination thereof. In some embodiments, the ferroelectric layers **132** are deposited along the sidewalls and bottom surfaces of the second trenches (**130** of FIG. 1J) using ALD, CVD (such as LPCVD, PECVD and HDP-CVD), another suitable technique, and/or a combination thereof.

(87) In some embodiments, the channel layers **134** are made of a semiconductor material such as indium gallium oxide (InGaO), indium gallium zirconium oxide (InGaZrO), indium gallium zinc oxide (IGZO), polysilicon, germanium, another suitable material, or a combination thereof. In some embodiments, the channel layers **134** are deposited over the ferroelectric layers **132** using ALD, CVD (such as LPCVD, PECVD and HDP-CVD), another suitable technique, and/or a combination thereof.

(88) In some embodiments, the insulating layers **136** are made of dielectric material such as silicon oxide (SiO.sub.2), silicon oxynitride (SiON), silicon nitride (SiN), silicon oxycarbide (SiOC), silicon carbon nitride (SiCN), silicon oxycarbonitride (SiOCN), aluminum oxide (Al.sub.2O.sub.3), zirconium dioxide (ZrO.sub.2), hafnium oxide (HfO.sub.2) yttrium oxide (Y.sub.2O.sub.3), lanthanum oxide (La.sub.2O.sub.3), another suitable material, and/or a combination thereof. In some embodiments, the insulating layers **136** are deposited over the channel layers **134** using ALD, CVD (such as LPCVD, PECVD and HDP-CVD), another suitable technique, and/or a combination thereof.

(89) In some embodiments, the capping layers **138** are made of dielectric material such as silicon oxide (SiO.sub.2), silicon oxynitride (SiON), silicon nitride (SiN), silicon oxycarbide (SiOC), silicon carbon nitride (SiCN), silicon oxycarbonitride (SiOCN), another suitable material, and/or a combination thereof. In some embodiments, the capping layers **138** are made of the same material as the first dielectric layers **104**, e.g., an oxide-based dielectric material (such as silicon oxide). In some embodiments, the capping layers **138** are deposited over the insulating layers **136** using ALD, CVD (such as LPCVD, PECVD and HDP-CVD), another suitable technique, and/or a combination thereof.

(90) After the material for the capping layers **138** is deposited, an etching back process is performed on the ferroelectric layers **132**, the channel layers **134**, the insulating layers **136**, and the capping layers **138** to open the bottoms of the second trenches **131**, so that the second trenches **131** extend to the substrate **102**, in accordance with some embodiments. The etching process may be an anisotropic etching process such as dry plasma etching, an isotropic etching process such as dry chemical etching, remote plasma etching or wet chemical etching, and/or a combination thereof.

(91) The etching back process also removes portions of the ferroelectric layers **132**, the channel layers **134**, the insulating layers **136** and the capping layers **138** formed above the upper surfaces of the strips **114'**, and further recesses portions of the ferroelectric layers **132**, the channel layers **134**, the insulating layers **136** and the capping layers **138** formed in the second trenches **131**, in accordance with some embodiments.

(92) In some embodiments, each of the ferroelectric layer **132**, the channel layer **134**, the insulating layer **136** and the capping layer **138** has a closed-loop profile, as shown in FIG. **1K-1**.

(93) FIGS. **1K-2** and **1K-3** are cross-sectional views taken along lines I-I and II-II shown in FIG. **1K-1**, in accordance with some embodiments. The channel layers **134** are formed along the substantially flat sidewalls **122B** of the first conductive lines **122**, as shown in FIGS. **1K**, **1K-1**, **1K-2** and **1K-3**, in accordance with some embodiments. Because the concave portions **122A** of the sidewalls of the first conductive lines **122** are covered by the second dielectric pillars **128**, the channel layers **134** are not formed along the concave portions **122A**, in accordance with some embodiments. Therefore, the channel layers **134** may also have flat profiles, thereby improving the performance of the resulting semiconductor memory device.

(94) FIGS. **1L** and **1L-1** illustrate the formation of fill layers **140**, in accordance with some embodiments.

(95) Fill layers **140** are formed over the semiconductor memory structure **100** to fill in the second trenches (**131** of FIG. **1K**), as shown in FIGS. **1L** and **1L-1**, in accordance with some embodiments. In some embodiments, the fill layers **140** are made of dielectric material such as silicon oxide (SiO₂), silicon oxynitride (SiON), silicon nitride (SiN), silicon oxycarbide (SiOC), silicon carbon nitride (SiCN), silicon oxycarbonitride (SiOCN), another suitable material, and/or a combination thereof. In some embodiments, the fill layers **140** are made of the same material as the first dielectric layers **104**, e.g., an oxide-based dielectric material (such as silicon oxide).

(96) In some embodiments, the fill layers **140** are formed by depositing a dielectric material to overfill the second trenches **131**, and then planarizing the dielectric material to remove the portion of the dielectric material formed above the strips. The deposition process may be ALD, CVD (such as LPCVD, PECVD, HDP-CVD, HARP, and FCVD), another suitable technique, and/or a combination thereof. The planarizing process may be an etching back process and/or chemical mechanical polishing (CMP).

(97) FIGS. **1M** and **1M-1** illustrate the formation of third through holes **142**, in accordance with some embodiments. FIGS. **1M-2** and **1M-3** are cross-sectional views taken along lines I-I and II-II shown in FIG. **1M-1**, in accordance with some embodiments.

(98) After the fill layers **140** are formed, a planarization process is performed on the semiconductor memory structure **100** until the ferroelectric layers **132**, the channel layers **134**, the insulating layers **136** and the capping layers **138** are exposed, in accordance some embodiments. The planarizing process may be an etching back process and/or chemical mechanical polishing (CMP).

(99) Third through holes **142** are then formed through the fill layers **140**, the capping layers **138**, the insulating layers **136**, as shown in FIGS. **1M**, **1M-1**, **1M-2** and **1M-3**, in accordance with some embodiments. In some embodiments, the closed-loop profiles of the insulating layer **136** and the capping layer **138** are cut off by the third through holes **142** while the closed-loop profiles of the ferroelectric layer **132** and the channel layer **134** remain intact (or continuous), as shown in FIG. **1M-1**. In some embodiments, the third through holes **142** expose the channel layers **134** and the substrate **102**, in accordance with some embodiments.

(100) In some embodiments, the formation of the third through holes **142** includes forming a patterned mask layer (not shown) having opening patterns corresponding to the third through holes **142** over the semiconductor memory structure **100**, and then etching the semiconductor memory structure **100** using the patterned mask layer to transfer the opening patterns into the fill layers **140**, the capping layers **138**, the insulating layers **136** until the substrate **102** is exposed. In some embodiments, the patterned mask layer is a patterned photoresist layer which is formed by a photolithography process. The etching process may be an anisotropic etching process such as dry plasma etching, an isotropic etching process such as dry chemical etching, remote plasma etching, wet chemical etching, and/or a combination thereof.

(101) In some alternative embodiments, the patterned mask layer is a patterned hard mask layer, which is formed by depositing a dielectric layer, forming a patterned photoresist layer over the dielectric layer, and etching the dielectric layer to transfer the opening patterns of the photoresist layer into the dielectric layer. The patterned mask layer may be removed during the etching process or by an additional process (such as etching, wet strip and/or ashing).

(102) The third through holes **142** are arranged in a row/column configuration, in accordance with some embodiments. For example, rows of third through holes **142** extend in the X direction, and columns of third through holes **142** extend in the Y direction. In some embodiments, the third through holes **142** are not formed at the positions where the second dielectric pillars **128** have been disposed. Although FIG. **1M-1** illustrates that nine third through holes **142** are formed between neighboring two second dielectric pillars **128**, the number of the third through holes **142** between neighboring two second dielectric pillars **128** is not limited thereto and may depend on design demand and/or performance consideration.

(103) FIGS. **1N** and **1N-1** illustrate the formation of second conductive lines **144**, in accordance with some embodiments. FIGS. **1N-2** and **1N-3** are cross-sectional views taken along lines I-I and II-II shown in FIG. **1N-1**, in accordance with some embodiments.

(104) Second conductive lines **144** are formed in the third through holes (**142** of FIG. **1M**), as shown in FIGS. **1N**, **1N-1**, **1N-2** and **1N-3**, in accordance with some embodiments. The second conductive lines **144** extend in the Z direction and are in contact with the channel layers **134** and the substrate **102**, in accordance with some embodiments. In some embodiments, the second conductive lines **144** are source/drain electrodes of memory transistors and function as source lines or bit lines of the resulting semiconductor memory device, such as NOR-type flash memory.

(105) In some embodiments, the second conductive lines **144** are made of one or more electrically conductive material, for example, tungsten (W), cobalt (Co), ruthenium (Ru), copper (Cu), aluminum (Al), nickel (Ni), titanium (Ti), tantalum (Ta), molybdenum (Mo), titanium nitride (TiN), tantalum (Ta), tantalum nitride (Ta₂N₃), titanium (Ti), cobalt tungsten (CoW), another suitable material, and/or a combination thereof.

(106) The formation of the second conductive lines **144** includes depositing conductive material for the second conductive lines **144** to overfill the third through holes (**142** of FIG. **1M**), and then planarizing the conductive material to remove the portion of the conductive material formed above the ferroelectric layers **132**, the channel layers **134**, the insulating layers **136** and the capping layers **138** until the ferroelectric layers **132**, the channel layers **134**, the insulating layers **136** and the capping layers **138** are exposed, in accordance with some embodiments. The deposition process may be ALD, CVD, PVD, e-beam evaporation, ECP, ELD, another suitable method, or a combination thereof. The planarizing process may be an etching back process and/or chemical mechanical polishing (CMP).

(107) In some embodiments, the first conductive lines **122**, the ferroelectric layers **132**, the channel layers **134**, and the second conductive lines **144** combine to form memory transistors (e.g., flash memory transistors), thereby functioning as memory cells. The memory cells are operable for data storage and arranged in the X, Y and Z directions to form a memory cell array, in accordance with some embodiments. In some embodiments, the second conductive lines **144** are electrically

connected to the peripheral circuits formed in the substrate **102**.

(108) Additional CMOS manufacturing processes may be formed, in accordance with some embodiments. For example, inter metal dielectric layer and metal layers and vias may be formed over the semiconductor memory structure **100**, in accordance with some embodiments.

(109) As described above, by forming the first and second dielectric pillars **110** and **128**, the risk of collapsing and/or wiggling of the strips **114** and **114'** may decrease. As a result, the manufacturing yield of the resulting semiconductor memory device may improve. In addition, by forming the sacrificial layers **116**, the first conductive lines **122** may be formed with substantially flat sidewalls, and the channel layers **134** formed thereon also have substantially flat profiles. As a result, the performance of the resulting semiconductor memory device may improve.

(110) FIG. **2** is a cross-sectional view of a semiconductor memory structure **200**, in accordance with some embodiments of the disclosure. The embodiments of the FIG. **2** are similar to the embodiments of the FIGS. **1N**, **1N-1**, **1N-2** and **1N-3**, except that FIG. **2** further illustrates that a semiconductor substrate, an interconnect structure over the semiconductor substrate, and a memory cell array over the semiconductor substrate.

(111) A semiconductor substrate **202** is provided, as shown in FIG. **2**, in accordance with some embodiments. In some embodiments, the semiconductor substrate **202** is a silicon substrate. In some embodiments, the semiconductor substrate **202** includes an elementary semiconductor such as germanium; a compound semiconductor such as gallium nitride (GaN), silicon carbide (SiC), gallium arsenide (GaAs), gallium phosphide (GaP), indium phosphide (InP), indium arsenide (InAs), and/or indium antimonide (InSb); an alloy semiconductor such as SiGe, GaAsP, AlInAs, AlGaAs, GaInAs, GaInP, and/or GaInAsP; or a combination thereof.

(112) Peripheral circuits **204** are formed on the semiconductor substrate **202**, as shown in FIG. **2**, in accordance with some embodiments. The peripheral circuits **204** may make up a control circuit for operating an array of memory cells formed vertically above. The peripheral circuits **204** may include, but is not limited to, voltage boost circuitry, page buffer circuitry, column decoder, row decoder, error correction circuitry, write assist circuitry, interface circuitry including for interfacing between types of memory cells, bus control circuitry, and the like.

(113) In some embodiments, the peripheral circuits **204** include MOSFETs such as p-type MOSFETs (P-MOSFET) or n-type MOSFETs (N-MOSFET). The MOSFET may be planar-type transistors, fin-type transistors (e.g., FinFETs), and/or another applicable transistor. In some embodiments, the peripheral circuits **204** are planar-type transistors. The peripheral circuits **204** each include a gate structure formed over the upper surface of the semiconductor substrate **202** and source/drain regions formed in or embedded at least partially in the semiconductor substrate **202** on opposite sides of the gate structure, in accordance with some embodiments.

(114) In some embodiments, the gate structure may include a gate dielectric layer and a gate electrode layer over the gate dielectric layer. In some embodiments, the gate dielectric layer includes an interfacial layer of dielectric material such as silicon oxide (SiO₂), hafnium silicon oxide (HfSiO), or silicon oxynitride (SiON). The interfacial layer may be formed using chemical oxidation, thermal oxidation, ALD, CVD, and/or another suitable method. In some embodiments, the gate dielectric layer includes high-K gate dielectric layer of high-K dielectric materials such as hafnium oxide (HfO₂), TiO₂, HfZrO, Ta₂O₃, HfSiO₄, ZrO₂, ZrSiO₂, LaO, AlO, ZrO, TiO, Ta₂O₅, Y₂O₃, SrTiO₃ (STO), BaTiO₃ (BTO), BaZrO, HfZrO, HfLaO, HfSiO, LaSiO, AlSiO, HfTaO, HfTiO, (Ba,Sr)TiO₃ (BST), Al₂O₃, Si₃N₄, oxynitrides (SiON), combinations thereof, or another suitable material. The high-K gate dielectric layer may be formed by ALD, physical vapor deposition (PVD), CVD, thermal oxidation, and/or another suitable method.

(115) In some embodiments, the gate electrode layer includes a conductive material, such as doped semiconductor, a metal, metal alloy, or metal silicide. In some embodiments, the gate electrode layer includes a single layer or alternatively a multi-layer structure. The gate electrode layer may be

formed of polysilicon, germanium, Ti, Ag, Al, TiAlN, TaC, TaCN, TaSiN, Mn, Zr, TiN, TaN, Ru, Mo, Al, WN, Cu, W, Re, Ir, Co, Ni, another suitable conductive material, or multi-layers thereof. The gate electrode layer may be formed by ALD, PVD, CVD, e-beam evaporation, or another suitable process.

(116) In some embodiments, the source/drain regions are regions of the substrate **302** suitably doped using an implantation process. In some embodiments, the source/drain regions are epitaxially grown source/drain regions using an epitaxial growth process.

(117) An interconnect structure is formed over the semiconductor substrate **202**, in accordance with some embodiments. The interconnect structure includes contacts **208**, metal layers **210** and vias **212** in one or more intermetal dielectric (IMD) layers **206**, as shown in FIG. 2, in accordance with some embodiments. The interconnect structure serves to electrically coupled the underlying peripheral circuits **204** with an overlying memory cell array, in accordance with some embodiments.

(118) The one or more intermetal dielectric layers **206** are made of one or more dielectric materials, such as silicon oxide (SiO₂), silicon nitride (SiN), silicon oxynitride (SiON), silicon carbon nitride (SiCN), oxygen-doped silicon carbide (SiC:O), oxygen-doped silicon carbon nitride (SiCN:O), silicon oxycarbide (SiOC), tetraethylorthosilicate (TEOS) oxide, un-doped silicate glass (USG), or doped silicon oxide such as borophosphosilicate glass (BPSG), fluoride-doped silicate glass (FSG), phosphosilicate glass (PSG), borosilicate glass (BSG), and/or another suitable dielectric material.

(119) The contacts **208** are formed in and/or through the one or more intermetal dielectric layers **206** and land on the gate structure of the peripheral circuits **204**, in accordance with some embodiments. In some alternative embodiments, the contacts **208** land on the source/drain regions. In some embodiments, the contacts **208** are made of one or more conductive materials, for example, cobalt (Co), nickel (Ni), tungsten (W), titanium (Ti), tantalum (Ta), copper (Cu), aluminum (Al), ruthenium (Ru), molybdenum (Mo), TiN, TaN, and/or a combination thereof.

(120) The metal layers **210** and the vias **212** are formed in and/or through the one or more intermetal dielectric layers **206**, in accordance with some embodiments. The metal layers **210** provide a horizontal routing for a signal or signals produced by the peripheral circuits **204**, in accordance with some embodiments. The vias **212** provide a vertical routing for a signal or signals produced by the peripheral circuit **204**, in accordance with some embodiments. In some embodiments, the metal layers **210** and the vias **212** are made of copper (Cu), cobalt (Co), ruthenium (Ru), molybdenum (Mo), tungsten (W), titanium (Ti), another suitable conductive material, an alloy thereof, nitrides of these materials, multi-layers thereof, and/or a combination thereof.

(121) Afterward, the steps as described above with respect to FIGS. 1A through 1N-1 are performed to form a memory cell array over the interconnect structure, as shown in FIG. 2, in accordance with some embodiments. In some embodiments, the second conductive lines **144** of the memory cell array are electrically connected to the peripheral circuits **204** through the interconnect structure. Although not illustrated in FIG. 2, the first conductive lines **122** may also be electrically connected to the peripheral circuits **204** through the interconnect structure. The semiconductor memory structure **200** is referred to as a peripheral circuit under memory array (PUA) device, in accordance with some embodiments.

(122) FIGS. 3A-3B are cross-sectional views illustrating the formation of a semiconductor memory structure **300** at various intermediate stages, in which FIGS. 3A and 3B are modifications of FIGS. 1H-2 and 1N-2, in accordance with some embodiments of the disclosure. FIGS. 3A and 3B correspond to line I-I shown in FIGS. 1H-1 and 1N-1, in accordance with some embodiments. The embodiments of the FIGS. 3A-3B are similar to the embodiments of the FIGS. 1A through 1N-3, except that the etching depths D51-D54 of the notches **119** vary as the level of the first conductive lines **122**.

(123) After the deposition of the barrier layer **124** and the metal bulk layer **126**, an etching back process is performed to remove portion of the barrier layer **124** and the metal bulk layer **126** outside the gaps **120**, thereby forming the first conductive lines **122**. In some embodiments, due to the characteristics of the etching process, portions of the first conductive lines **122** exposed from the second through holes **118** may be laterally recessed, e.g., by maximum depths **D51-D54**, as shown in FIG. **3A**. In some embodiments, the depths **D51-D54** decrease as the position of the first conductive lines **122** descends (or the level of the first conductive lines **122** decreases).

(124) The steps described above with respect to FIGS. **1I** through **1N-3** are performed, thereby forming the second dielectric pillars **128**, the ferroelectric layers **132**, the channel layers **134**, the insulating layers **136**, the capping layers **138**, the fill layers **140** and the second conductive lines **144**, as shown in FIG. **3B**, in accordance with some embodiments. The second dielectric pillars **128** includes protruding portions filled in the notches **119**, in accordance with some embodiments. In some embodiments, the dimensions of the protruding portions (in the X direction) decrease as the position of the first conductive lines **122** descends (or the level of the first conductive lines **122** decreases).

(125) FIGS. **4A-4B** are cross-sectional views illustrating the formation of a semiconductor memory structure **400** at various intermediate stages, in which FIGS. **4A** and **4B** are modifications of FIGS. **1H-2** and **1N-2**, in accordance with some embodiments of the disclosure. FIGS. **4A** and **4B** correspond to line I-I shown in FIGS. **1H-1** and **1N-1**, in accordance with some embodiments. The embodiments of the FIGS. **4A-4B** are similar to the embodiments of the FIGS. **1A** through **1N-3**, except that the sidewalls of the first conductive lines **122** have no concave portion.

(126) After the deposition of the barrier layer **124** and the metal bulk layer **126**, an etching back process is performed to remove portions of the barrier layer **124** and the metal bulk layer **126** outside the gaps **120**, thereby forming the first conductive lines **122**. In some embodiments, due to the characteristics of the etching process, the sidewalls of the first conductive lines **122** facing the second through holes **118** are also substantially flat.

(127) The steps described above with respect to FIGS. **1I** through **1N-3** are performed, thereby forming the second dielectric pillars **128**, the ferroelectric layers **132**, the channel layers **134**, the insulating layers **136**, the capping layers **138**, the fill layers **140** and the second conductive lines **144**, as shown in FIG. **4B**, in accordance with some embodiments.

(128) As described above, the method for forming a semiconductor memory structure includes forming a stack including first dielectric layers **104** and second dielectric layers **106** alternately arranged, forming first dielectric pillars **110** through the stack, and etching the stack to form a plurality of first trenches **112** and strips **114** between the first trenches **112**. The first dielectric pillars **110** may support the strips **114**, thereby decreasing the risk of collapsing and/or wiggling of the strips **114**, in accordance with some embodiments. Therefore, the manufacturing yield of the resulting semiconductor memory device may improve.

(129) In addition, the method also includes forming sacrificial layers **116** in the first trenches **112**, replacing the second dielectric layers **106** with first conductive lines **122**, and forming channel layers **134** along strips **114'** including the first conductive lines **122** and the first dielectric layers **104**. As a result, the first conductive lines **122** may be formed with substantially flat sidewalls, and the channel layers **134** formed thereon also have substantially flat profiles. Therefore, the performance of the resulting semiconductor memory device may improve.

(130) Embodiments of a semiconductor memory structure may be provided. The semiconductor memory structure may include a strip including dielectric layers and first conductive lines alternatively stacked, a second conductive line vertically extending along a first side of the strip, and a dielectric pillar vertically extending along a second side of the strip. The dielectric pillars may support the strips, thereby decreasing the risk of collapsing and/or wiggling of the strips. Therefore, the manufacturing yield of the resulting semiconductor memory device may improve.

(131) In some embodiments, a method for forming a semiconductor memory structure is provided.

The method includes forming a stack over a substrate, and the stack includes first dielectric layers and second dielectric layers vertically alternately arranged. The method also includes forming first dielectric pillars through the stack, and etching the stack to form first trenches. Sidewalls of the first dielectric pillars are exposed from the first trenches. The method also includes removing the first dielectric pillars to form through holes, removing the second dielectric layers of the stack to form gaps between the first dielectric layers, and forming first conductive lines in the gaps.

(132) In some embodiments, a semiconductor memory structure is provided. The semiconductor memory structure includes a strip, and the strip includes dielectric layers and first conductive lines alternatively stacked over a substrate. The semiconductor memory structure also includes a second conductive line vertically extending along a first side of the strip, a channel layer sandwiched between the strip and the second conductive line, and a dielectric pillar vertically extending along a second side of the strip that is opposite the first side of the strip.

(133) In some embodiments, a semiconductor memory structure is provided. The semiconductor memory structure includes a first word line and a second word line laterally spaced apart from the first word line, a first channel layer and a second channel layer between the first word line and the second word line, and a dielectric pillar between the first word line and the second word line and between the first channel layer and the second channel layer. The dielectric pillar includes a first protruding portion extending into the first word line and a second protruding portion extending into the second word line.

(134) The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A method for forming a semiconductor memory structure, comprising: forming a stack over a substrate, the stack comprising first dielectric layers and second dielectric layers vertically alternately arranged; forming first dielectric pillars through the stack; etching the stack to form first trenches, wherein sidewalls of the first dielectric pillars are exposed from the first trenches; removing the first dielectric pillars to form through holes; removing the second dielectric layers of the stack to form gaps between the first dielectric layers; and forming first conductive lines in the gaps.
2. The method for forming the semiconductor memory structure as claimed in claim 1, further comprising: forming sacrificial layers in the first trenches before removing the first dielectric pillars; and removing the sacrificial layers after forming the first conductive lines in the gaps to form second trenches.
3. The method for forming the semiconductor memory structure as claimed in claim 2, further comprising: forming a ferroelectric layer in the second trenches; forming a channel layer over the ferroelectric layer in the second trenches; and forming an insulating layer over the channel layer in the second trenches.
4. The method for forming the semiconductor memory structure as claimed in claim 3, further comprising: forming second conductive lines vertically penetrating through the insulating layer.
5. The method for forming the semiconductor memory structure as claimed in claim 1, wherein removing the second dielectric layers of the stack comprises introducing an etchant into the through holes to laterally etch the second dielectric layers of the stack.

6. The method for forming the semiconductor memory structure as claimed in claim 1, further comprising: filling the through holes with second dielectric pillars.
7. The method for forming the semiconductor memory structure as claimed in claim 1, wherein forming the first conductive lines in the gaps comprises: forming a barrier layer to partially fill the gaps; forming a metal bulk layer to overfill remaining portions of the gaps; and etching back the barrier layer and the metal bulk layer.
8. The method for forming the semiconductor memory structure as claimed in claim 1, wherein a sidewall of one of the first conductive lines has a portion that is exposed from one of the through holes, and the portion of the sidewall is concave.
9. The method for forming the semiconductor memory structure as claimed in claim 1, wherein the stack is etched to form strips between the first trenches, and the first dielectric pillars are in contact with sidewalls of the strips.
10. A method for forming a semiconductor structure, comprising: forming a stack comprising a plurality of first dielectric material layers interleaved with a plurality of second dielectric material layers; etching the stack to form a plurality of strips separated by trenches, the plurality of strips respectively comprising first dielectric material layer segments interleaved with second dielectric material layer segments; replacing the second dielectric material layer segments with first conductive lines; forming second dielectric pillars within the trenches after forming the first conductive lines, the second dielectric pillars being separated by second trenches; forming a ferroelectric layer, a channel layer, and an insulating layer within the second trenches; and forming one or more second conductive lines within the insulating layer.
11. The method of claim 10, further comprising: forming a sacrificial material within the trenches; removing the second dielectric material layer segments, after forming the sacrificial material within the trenches, to form gaps between neighboring ones of the first dielectric material layer segments; and forming the first conductive lines within the gaps.
12. The method of claim 11, further comprising: removing the sacrificial material, after forming the second dielectric pillars.
13. The method of claim 11, wherein one or more second through holes vertically extend through the sacrificial material and expose sidewalls of the second dielectric material layer segments within the plurality of strips.
14. The method of claim 13, further comprising: forming first dielectric pillars within the trenches; forming the sacrificial material within the trenches and along sidewalls of the first dielectric pillars; and removing the first dielectric pillars to form the one or more second through holes.
15. The method of claim 14, further comprising: forming the second dielectric pillars within the one or more second through holes after removing the second dielectric material layer segments.
16. The method of claim 10, wherein the first conductive lines have sidewalls having concave portions and the second dielectric pillars respective include protruding portions that mate with the concave portions of the sidewalls of the first conductive lines.
17. A method for forming a semiconductor structure, comprising: forming a stack over a substrate, the stack comprising a plurality of first dielectric layers interleaved with a plurality of second dielectric layers; etching the stack to form a plurality of trenches; forming a sacrificial material within the plurality of trenches; forming second through holes that extend through the sacrificial material and expose sides of the plurality of second dielectric layers; introducing an etchant into the second through holes to remove the plurality of second dielectric layers and form a first plurality of gaps between neighboring ones of the plurality of first dielectric layers; forming first conductive lines within the first plurality of gaps; forming second dielectric pillars within the second through holes; removing the sacrificial material to form a second trenches; forming a ferroelectric layer, a channel layer, and an insulating layer within the second trenches, the ferroelectric layer arranged along sidewalls of the plurality of first dielectric layers, the first conductive lines, and the second dielectric pillars; and forming one or more second conductive lines along a sidewall of the

insulating layer.

18. The method of claim 17, wherein the channel layer is arranged along opposing sides of the one or more second conductive lines.

19. The method of claim 17, wherein the channel layer and the ferroelectric layer continuously extend in a closed loop surrounding the insulating layer and the one or more second conductive lines.

20. The method of claim 17, further comprising: forming a capping layer on the insulating layer and within the second trenches; forming a fill layer on the capping layer and within the second trenches; removing parts of the insulating layer, the capping layer, and the fill layer to form third through holes exposing sidewalls of the channel layer; and forming the one or more second conductive lines within the third through holes.
